

Scale of the problem

8,800

ADA Title III federal lawsuits, 2024

\$236,451

DOJ max penalty per repeat violation (2025)

~73%

U.S. buildings fail $\geq$ 1 ADA standard (indus

~70M

U.S. adults with a disability (CDC 2022)

32%

of buyers ask about ADA at inspection

\$800-\$50K+

per CASp audit; 2-5 days on-site

Hundreds

of buildings on the TAMU College Station c

What STAR is

Distributed robot:

- Raspberry Pi 5 + custom RX72N PCB
- SPI 10 Mbps + HARQ + FEC + nanob
- 143 ROS2 tests passing

Sensors (all wired, all tested):

- RPLiDAR C1 (360, 10 Hz, 12 m)
- Waveshare IMX219-83 stereo (dual 8 MP, 60 mm baseline)
- BNO055 IMU on chassis
- ICM20948 IMU on camera
- 4x HC-SR04 ultrasonic
- DS18B20 + BMP280

Motor drivetrain:

- 4x DFRobot FIT0520 gearmotors
- 4x TI DRV8263H H-bridges
- GPTW 32-bit PWM @ 20 kHz
- MTU/TPU quadrature encoders
- 250 Hz PID on RX72N

Software: slam_toolbox async + robot_localization EKF + Nav2 + m-explore-ros2 + compliance engine

Total BOM: under \$2,000.

7 ADA checks - status

1. Ramp slope $> 1:12$ **IMPLEMENTED**
ADA 405.2

2. Trip hazard $> 0.25"$ **STRETCH**
ADA 303

3. Path width $< 36"$ **STRETCH**
ADA 403.5

4. Ramp width $< 36"$ **architected**
ADA 405.5

5. Ramp landing $< 5 \times 5'$ **architected**
ADA 405.7

6. Door clear $< 32"$ **architected**
ADA 404.2.3

7. Door threshold $> 0.5"$ **architected**
ADA 404.2.5

Implementation code in compliance-engine/

Validation & ask

Platform validation (done):

143 / 143 ROS2 tests passing
slam_toolbox /map at 0.5 Hz
Full TF chain active
Nav2 autonomous goals run
Frontier exploration verified
SPI 1.6% utilization at 10 Mbps

Ramp-slope check validation:

Wixey WR300 (+/- 0.1 deg) truth
BNO055 vs. LiDAR normal, +/- 0.5 deg gate
On-campus ramp, n=5-30 sessions
Live numbers: validation_log.csv

At TAMU scale:

5,200 acres, hundreds of buildings
Manual audit = multi-million-dollar cost
STAR = amortized hardware cost,
re-runnable monthly

github.com/Locked-Inc/STAR

Pilot building partnerships welcome.